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CO2 Performance Ladder Level 5

Handbook

Annual Report 2024

CO2 Performance Ladder Level 5

Handbook

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1 Introduction

Tijhuis Ingenieurs takes responsibility in the field of sustainable business. Our goal in January 2021 was to certify ourselves under the CO2 Performance Ladder Level 4. This goal was achieved in April 2021. In April 2022, during the renewal audit, we decided that we were ready to achieve Level 5. The certificate for the CO2 Performance Ladder Level 5 was obtained on 26 April 2022. Since then, our sustainability ambitions have gained real momentum, and we achieve excellent results in sustainability every year. This document describes all our steps in the field of sustainability.

Step-by-Step Plan

To implement the CO2 Performance Ladder, we used the document Handbook CO2 Performance Ladder version 3.1 as a guideline. We have completed the following steps:

forming a project team;

calculating the CO2 footprint;

formulating measures;

communication and assurance.

Reader's Guide

The result of these steps is described in this document. In Chapter 2, we first provide an explanation of our company and the project team. Chapter 3 then calculates the baseline situation: our CO2 footprint. Chapter 4 contains the policy statement of the management board, and addresses our quantitative objectives, our reduction targets and the steps we are taking to curb our emissions. Chapter 5 presents, in tabular form, all measures we are going to take and have taken. We also try to influence our supply chain in various ways, which is described in Chapter 6. How we communicate about all of this within the company and externally can be found in Chapter 7. Finally, Chapter 8 contains the assurance.

In this document, the CO2 handbook of 2023 has been adapted, and we look back at the past year (2024), the current situation, and the objectives for next year (2025).

2 Sustainability at Tijhuis Ingenieurs

2.1 The company

We are an independent and leading engineering consultancy in the field of regional water management in the Netherlands. Our fields of work are: water bed and dredging, water quality and ecology, banks and structures, and GIS and data. Within these fields we provide a complete service, from data acquisition, drafting plans and writing specifications through to supervising implementation. We are pioneers in the digital acquisition, processing, analysis and presentation of data using GIS applications.

With 65 colleagues, we have extensive knowledge of the planning phase, the preparatory research phase and the practical implementation phase of dredging works. From our offices in Hoorn, Amersfoort and Joure, we undertake projects throughout the country. 9 certified field workers and 4 field workers in training (protocol 2003) carry out sampling and measurements of the water bed and the water on a daily basis.

We carry out our work from the three offices in Hoorn, Amersfoort and Joure. These three offices fall under the company 'Tijhuis Ingenieurs BV'. Administratively, Tijhuis Ingenieurs BV falls successively under 'Tijhuis Groep BV' and 'Tijhuis Beheer BV'. Only Tijhuis Ingenieurs falls under the CO2 certificate. All employees of 'Tijhuis Groep BV' and 'Tijhuis Beheer BV' work in the offices of 'Tijhuis Ingenieurs BV', so they are automatically included with the same footprint. In addition, 'Tijhuis Groep BV' and 'Tijhuis Beheer BV' do not carry out any projects. Within the organisational boundaries, Tijhuis Ingenieurs BV has no parent company or subsidiaries that are included in the implementation process of the CO2 Performance Ladder. The organisation of Tijhuis Ingenieurs is shown in the figure below.

Figure 2.1 Organisation chart

We are included in the list of Kwalibo-registered soil intermediaries. Tijhuis Ingenieurs BV holds the following relevant certificates:

- ISO 9001: 2015;
- VCA*: 2017/6.0;
- BRL 2000: protocol 2003;
- BRL 6000: protocol 6003;
- CO2 Performance Ladder level 5.

2.2 The sustainability project team

To further shape the sustainable company, we have formed a broadly oriented project team under the leadership of the KAM coordinator. The project organisation is further detailed in the organisational chart below.

Figure 2.2 Organisational chart of the CO2 project team

The project team has made the emission factors transparent. Subsequently, the reduction measures were formulated and, after approval, signed by the board of directors. The project team is responsible for the implementation and execution of all measures. Monitoring and adjustment are carried out by the KAM coordinator.

2.3 Sustainable advice in and around water

In 2023, we tightened our sustainability targets. As part of this, PHI Factory, in collaboration with colleagues, drew up a roadmap in which a target has been set for each of the themes below:

Water and water bed quality

Ecology

Material

Energy

In Your Power

Our Overarching Goal

Our overarching goal is as follows: 'Tijhuis Ingenieurs will be the leading expert in sustainable regional water in the Netherlands by 2030.' In addition, we have set ourselves the objective of becoming climate neutral by 2030. Our online management system for professionals in the water bed, 'WIT', is the means by which we see a good chance of achieving a large part of our goals.

From Ambition to Practice: The PHI Factory Trajectory

Since 2020, we have been committed to sustainability. There are solar panels on the roof, as many electric cars and buses as possible are included in the vehicle fleet, and all field teams use electric outboard motors. Great strides have already been made in sustainability, and there are plenty of ideas. In 2023, we made our sustainability plans concrete and structured them together with PHI Factory. Through this, we aim to become even more sustainable in the future, both within our projects and within our organisation. In 2026, we will start a new trajectory with PHI Factory to further improve our sustainability approach.

Action Plan

PHI Factory has based the structured approach to making Tijhuis Ingenieurs more sustainable on the following steps:

An internal analysis to gain a clear picture of the current situation at Tijhuis Ingenieurs. This handbook and other documents prepared for achieving CO2 Performance Ladder level 5 have been used for this. This was then translated into a powerfully worded ambition and strategy.

Building on the first step, it was determined where opportunities lie within Tijhuis Ingenieurs for achieving the ambitions and objectives. In addition, it was determined which components have priority and how we can make our progress measurable.

Ultimately, these steps led to the creation of a visual roadmap and the drafting of an implementation programme setting out the sustainability ambition and objectives.

Outcome

With PHI Factory, we have created tools that everyone within the company can get started with. In this way, we can make Tijhuis Ingenieurs even better at sustainability and, in addition, fulfil a pioneering role within our sector.

3 Our CO2 Footprint

In the process towards certification for the CO2 Performance Ladder Level 4, we calculated our CO2 footprint for the first time based on the 2020 data. We then began reporting our footprint every six months for our Level 5 certification. This chapter elaborates on the justification for this.

3.1 CO2 Footprint versus Company Type

To determine which category Tijhuis Ingenieurs falls into, we calculated our CO2 footprint. This amounts to 149 tonnes per year (scope 1 and 2 emissions in 2024). Based on this value, we have been classified in the 'small company' category (see the reference table below).

Table 3.1 CO2 Emissions per Organisation Size

| Category | Maximum scope 1&2 emissions in tonnes CO2 per year | Works and supplies | Services | Construction sites and production locations | Offices and business premises |

|---|---|---|---|---|---|

| Small | | 2,000 | 500 | 500 | |

| Medium | | 10,000 | 2,500 | 2,500 | |

| Large | | >10,000 | >2,500 | >2,500 | |

Scope 1 and 2

We calculate emissions using CO2 emission factors from www.co2emissiefactoren.nl. We make all data transparent in an Excel spreadsheet, after which the data is automatically converted into emissions in tonnes of CO2 using the emission factors. The figure below shows our annual emissions from 2019 onwards. Chapter 3 also presents emissions per half-year.

Figure 3.1: Multi-year chart of scope 1 and 2 emissions

In 2019, total CO2 emissions for scope 1 and 2 amounted to 193 tonnes. In 2024, this was 149 tonnes of CO2, a reduction of approximately 23% compared with the reference year.

Scope 3

Besides our own emissions, we also indirectly emit CO2, for example in the dredging works we have prepared, through the hiring of field workers and through having sample analyses carried out. For scope 3, we have examined the sources with the greatest emissions. The scope 3 emissions amount to 433 tonnes for 2019 and 670 tonnes for 2024.

We have carried out an analysis of the most significant emissions. This analysis showed that we have the greatest influence on the PMC activity: 'Advice: Transport of dredged material by axle'. Our ambition is to reduce emissions to the level of our scope 1 & 2 footprint during the supervisory works; see chapter 4 for this.

3.2 Scopes

The CO2 footprint has been calculated according to scopes 1, 2 and 3 as set out in the SKAO handbook, version 3.1. The scopes are understood as follows (see also Figure 3.1):

● Scope 1 (direct emissions): emissions released by installations owned or controlled by the organisation, such as emissions from our own gas use (for example gas boilers, combined heat and power installations and furnaces) and our own vehicle fleet.

● Scope 2 (indirect emissions): emissions arising from the generation of electricity, heat, cooling and steam in installations that do not belong to our own organisation but that are used by the organisation, such as the emissions released when electricity is generated in power plants.

● Scope 3 (other indirect emissions): emissions arising as a result of the organisation's activities but originating from sources that are not owned or managed by the organisation. Examples are emissions from the production of purchased materials (upstream) and the use of the work, project, service or delivery offered/sold by the organisation (downstream).

Figure 3.2 Scope diagram

Organisational Boundary within Tijhuis Ingenieurs BV

The CO2 footprint is based on the consumption of energy, use of materials and transport of the branches in Hoorn, Amersfoort and Joure. The emission-causing factors are:

energy consumption of the offices and the garage units (Hoorn and Nieuwegein);

fuel for leased cars and vans (including field work vans);

declared kilometres for work-related travel in private cars;

energy consumption during fieldwork;

waste from offices.

Purchased Emission Factors of Tijhuis Ingenieurs BV

When compiling the CO2 footprint, account has been taken of the emission factors within the company itself. Tijhuis Ingenieurs carries out work for clients on a project basis. Most activities are performed by Tijhuis Ingenieurs itself. Where capacity is insufficient, external resources are engaged for fieldwork, such as self-employed contractors D. Holtjer and J. Borst, Ingenieursbureau Land, Heijdra Milieuservices bv and Waders Milieu bv. Purchased services fall under scope 3 emissions. The volume of contracted services varies considerably from year to year. By estimating the number of kilometres driven by contracted fieldworkers, we can nonetheless include the CO2 emissions within scope 3. External emission factors of this kind are documented as far as possible, so that opportunities for CO2 reduction can also be identified here.

3.3 CO2 Emission Data

From 2019 onwards, CO2 emissions have been calculated every six months. To calculate our emissions and, ultimately, the effect of the measures, we used the Milieubarometer of the Stichting Stimular up to and including 2022. Since 2023, we determine our emissions by means of an in-house overview, in which all required figures are multiplied by CO2 emission conversion factors available at: https://www.co2emissiefactoren.nl/.

The first CO2 footprint was determined in the course of 2020. In 2020, no reduction measures had yet been implemented within the organisation. At the beginning of 2021, solar panels were installed on the roof of the Amersfoort office as the first reduction measure. In addition, a great deal was learned from digital working during the pandemic. Digital working contributes to reducing CO2 emissions in the coming years. Furthermore, the 'Leaseplan Tijhuis Ingenieurs' was launched in January 2022. This is a first step towards making electric driving the norm. In 2023, additional solar panels were installed in Amersfoort and electric pool cars were purchased, so that colleagues need to use private cars less. In 2024 and early 2025, most of the lease cars were replaced by an electric alternative. In addition, the first electric field work van was put into use in 2025.

Data Sources

The data have been obtained in the following way:

Internal administration: The input data have been supplied from our internal administration. Overviews, accounts and invoices can be quickly made transparent here.

Leasing company reports: Use has been made of the reports prepared by the leasing companies for the lease cars and field work vans.

Colleague interviews: Other emission factors have been identified through direct contact with colleagues, such as sustainable choices made within the organisation.

Selection of Data and Accuracy

General

To determine which components are relevant for calculating our footprint, we have used the Milieubarometer. We have selected the factors that cause the most emissions and which we can influence. These we can monitor, and for the reduction of these components we can set realistic targets. These 'key figures' consist mainly of fuel consumption (of employees) and fixed costs such as electricity and gas consumption.

Fuel consumption and electricity for the vehicle fleet

The key figures for fuel consumption from 2023 and 2024 (approx. 20 vehicles) are accurate. Within Tijhuis Ingenieurs, business travel and commuting are by far the largest sources of emissions. We can request an annual report of the fuel and electricity consumed from the leasing companies (Hitachi Mobility Service and Greenchoice for the electric cars). In this way we have an accurate key figure for determining our CO2 footprint.

Electricity, gas and drinking water at offices

The electricity, gas and water consumption of our offices has been visible since 2020 by checking meter readings. Therefore, 2020 has been used as the zero situation for electricity and gas consumption. In Amersfoort we do not have a separate gas meter; because we use 40% of the building, we also calculate with 40% of the gas consumption.

In Joure we rent only two workstations. We calculate the consumption there on the basis of the consumption at our other offices. We do this by dividing the consumption by the number of FTEs in Joure and multiplying it by the number of FTEs in the other two offices.

Solar panels

In Amersfoort, the yield of the solar panels is recorded with an app, to which our landlord has given us access. In Hoorn, an estimate of the yield of the solar panels has been made on the basis of the number of panels and the estimated energy yield per solar panel. Unfortunately, we have no insight in Hoorn because the solar panels are owned by our landlord and we have no access to this data.

Wastewater

To calculate the volume of wastewater at our offices, we multiply the number of FTE indoor employees by publicly available benchmark figures. At present we assume 28 m^3 of wastewater per indoor employee per year.

Sample analyses and packaging transport

To calculate our emissions from our outsourcing to Eurofins, we request their CO2 emissions and turnover from Eurofins. We calculate our emissions on the basis of our share of Eurofins' turnover.

Hired-in fieldwork

To determine how much CO2 emission is caused by fieldworkers we hire, we work with the average distance and emissions that a fieldwork vehicle at Tijhuis Ingenieurs covers and emits per day. We multiply this emission by the number of days the hired fieldworker has worked for us. On the basis of invoices from hired fieldworkers, we determine how many days they have worked for us.

Transport of dredged material

The calculation of the emissions from transporting dredged material can be found in our chain analysis. We use the unit figures of co2emissiefactoren.nl, with which we calculate the emissions of

calculate the m³ of dredged material dispersed and transported. Our tender team provides this information from their project documentation.

Specification of CO2 emissions by project

At least 78% of the scope 1 & 2 CO2 emissions is related to transport for the projects we carry out, namely:

Transport to and from project locations. This mainly concerns the kilometres driven by fieldwork teams in vans, including equipment (fieldwork supplies). It also covers the business kilometres claimed for travel to projects by advisers and project managers.

Outboard motors. Fieldworkers carry out their activities using a small motorboat.

For projects based on an award advantage for the CO2 Performance Ladder, emissions are allocated using the financial allocation method (cost allocation).

3.4 Calculation of CO2 footprint for previous years

To calculate the CO2 footprint, the emissions data up to and including 2023 were multiplied by the emission factors set out at Factoren - Elektriciteit | CO2-emissiefactoren. In the first years this was done using the environmental barometer; later we switched to our own method in Excel. The total emissions were then calculated in tonnes of CO2 per year (scope 1 & 2). Appendix 1b contains the tables per year, arranged from newest to oldest.

3.5 CO2 footprint 2024

To calculate the CO2 footprint, the emissions data were entered in Excel and multiplied by the prevailing emission factors. The total emissions were then calculated in tonnes of CO2 per year (scope 1 & 2). Table 3.2 quantifies the emissions per category. The full footprint calculation is included in Appendix 1. Figure 3.3 shows the distribution of each category relative to total emissions, in both tonnes and percentages.

Table 3.2 CO2 emissions by category 2024 (scope 1 & 2)

Category	Emissions (tonnes CO2)	1st half 2024	2024
Scope 1			
Mobile machinery	0,17	0,43	
Fuel and heating, offices	5,4	11,3	
Fleet fuel use	60,4	116,2	
Scope 2			
Electricity*	11,8	21,3	
Scope 3			
Business travel	10,2	20,8	
Total	87,97	170	

*Including charging of electric cars

Figure 3.3 Tonnes of CO2 emissions 2024 by category (scope 1 & 2)

This shows the following:

● In 2024, the total CO2 emissions for scope 1 & 2 were 149 tonnes. Of this, 78.6 tonnes came from the first half of the year.

● Business travel produced emissions of 91 tonnes per year. At 78%, this is the largest emission source in our CO2 footprint (scope 1 & 2). This percentage is slightly lower than last year. We have also switched to electric driving on a much larger scale than in previous years. Our emissions from the electricity consumption of the cars have actually decreased thanks to the use of new charging cards with Dutch green electricity. Electricity consumption for the premises has not increased significantly, because more solar panels have been in use at Amersfoort since 2023. This effect is clearly visible in the overview of solar power generated.

3.6 Scope 3

To calculate the CO2 footprint for scope 3, the emissions data was entered into our Excel file. The total emissions in tonnes of CO2 were then calculated for 2019 through 2024. The emission sources consist mainly of the transport of dredged material, hired fieldwork and indirect emissions from sample analyses. The following table and figure quantify the emissions per category. The full footprint calculation is included in Appendix 1.

Table 3.3 CO2 emissions by category up to and including 2022 (scope 3)

| Category | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |

|---|---|---|---|---|---|---|

| Emissions (tonnes CO2) | | | | | | |

| Hired fieldwork | 8.95 | 1.54 | 1.46 | 0.69 | 11 | 20.2 |

| Sample analyses | 426 | 450 | 270 | 649 | 803 | 560 |

| Transport of dredged material | n/a# | n/a# | n/a# | n/a# | 1335 | 1685 |

| Commuting | n/a* | n/a* | n/a* | 10 | 20 | 20.8 |

| Other | 0 | 0 | 0.5 | 1.23 | 1.4 | 0.9 |

| Total | 435 | 451 | 272 | 651 | 2170 | 2287 |

*Commuting could not yet be calculated separately from business travel in these years.

Transport of dredged material was not yet traceable in these years.

Figure 3.4 Scope 3 emissions (tonnes per year)

This shows the following:

The total scope 3 CO2 emissions were 435 tonnes in 2019, 451 tonnes in 2020, 272 tonnes in 2021, 651 tonnes in 2022, 2150 tonnes in 2023 and 2287 in 2024.

Compared with 2022, emissions in 2023 increased. This is partly explained by the fact that more work was carried out in 2023. In addition, since 2023 the transport of dredged sediment has been included in scope 3.

The largest emission source is the transport of dredged sediment. In our projects we always try to minimise transport distances, but transport cannot always be avoided.

Another major emission source is the analysis of samples, mainly due to the drying of sample material.

The amount of emissions from hired-in fieldwork has increased sharply in 2024 compared with all previous years.

3.7 CO2 footprint related to turnover

Next, the total CO2 footprint is related to turnover. Table 3.4 provides a summary of this. Emissions are expressed in kilograms per €1,000 of turnover. Because turnover has changed considerably over the years, this is the best way to quantify and demonstrate reduction. The targets for the coming years are explained in §4.2.

Table 3.4 CO2 emissions per turnover

Year	Scope 1 and 2 emissions (t CO2)	Scope 1 and 2 emissions (kg CO2/€1,000)	Target	Result (%) Reduction	Result (%) Corrected for turnover
2019	189,2	47	Reference	-	-
2020	135,8	37	5.0%	28%	22%
2021	138,4	30	7.5%	27%	36%
2022	140	29	20.5%	21%	38%
2023	158	21	23%	33%	56%
2024	149	20	25,5	25,5	58%

3.8 Analysis of energy consumption

Figures 3.5 and 3.6 show the distribution between different emission sources for 2019 and 2024. 2019 is the baseline against which each new year is compared. In 2024, the ratio between the various emission sources remained almost the same as in 2019. The figures show that business travel is still by far the largest emission source within scope 1 and 2. For scope 3, sample analyses are the largest emission source (99.8%).

Figure 3.5 Energy consumption per emission source, scope 1 and 2, in 2019

Figure 3.6 Energy consumption per emission source, scope 1 and 2, in 2024

3.9 Emissions per scope

The emission sources have been allocated to the three scopes described in section 3.1. Annex 1 calculates the CO2 equivalent per scope. The table below summarises this.

Table 3.5 CO2 equivalent per scope

Scope | CO2 equivalent (tonnes CO2) 2019 | 2024

1 Direct emissions | 147 | 128

2 Indirect emissions | 27,2 | 21,3

3 Other indirect emissions | 433 | 2287

The total emissions in scope 1 are 128 tonnes per year in 2024 (19 tonnes less than in 2019). The main emission source is fuel consumption of the field vehicles, with emissions of 104 tonnes per year.

Scope 2 has total emissions of 21,3 tonnes per year in 2024 (almost 6 tonnes less than in 2019). The main emission sources are the purchased electricity for office premises and the charging of electric cars. This emission is expected to decrease through the use of Greenchoice charging cards in 2025.

Scope 3 has total emissions of 2287 tonnes per year, making it by far the highest. The analysis (including drying) of sample material has a major impact on emissions (560 tonnes). This emission is considerably higher than in 2019, because more sampling and analyses were carried out. In addition, transport of dredged spoil has been included in our scope 3 since 2023, with emissions amounting to 1685 tonnes in 2024.

4 CO2 Reduction Policy

4.1 Management Policy Statement

At Tijhuis Ingenieurs, we have in recent years focused increasingly on sustainability, so that we can make ever more sustainable choices. Part of this is reducing our CO2 emissions in order to achieve certification at level 5 of the CO2 Performance Ladder. Our objective is to reduce our emissions in scope 1 & 2 by at least 25.5% in 2024 compared with 2019. Each year we set the new reduction target for the coming year in the management review: the guideline figure is a 2.5% reduction compared with the previous year.

We have chosen to achieve level 5 of the CO2 Performance Ladder as a target because we want to start ambitiously with reducing our CO2 footprint. Since 2019 we have monitored more precisely and can identify more opportunities for CO2 reduction, energy saving and sustainability. During our quarterly meetings (company-wide), CO2 reduction and sustainability are a standing topic, so that all colleagues continue to pay attention to the subject. In addition, since 2023 we have had a sustainability team.

The whole company must remain informed about our CO2 footprint. It is the task of the entire organisation to reduce it. The reduction must be traceable to the measures taken that are included in this manual. For reduction, we first focus on our largest emission sources, as established in the research strategy:

Scope 1: Light commercial vehicles (fieldwork vans)

Scope 2: Purchased electricity and business travel

Scope 3: Advisory services

Important steps we want to take in the coming years include, for example, making our vehicle fleet sustainable as quickly as possible. Every new lease car will be electric, and our electric cars are already charged using charge cards providing guaranteed Dutch green electricity. To reduce traffic from privately owned petrol cars, since June 2023 we have had two electric pool cars in our fleet. Until 2024 we searched for good electric alternatives to the fieldwork vans, so that we could purchase them as soon as possible. In early 2025, an electric fieldwork van was delivered, with which a successful pilot was launched. At the end of 2025 and the beginning of 2026, several vans will be replaced with an electric alternative. In addition, since 2023 we have had the following ambitious objective: 'by 2030, Tijhuis Ingenieurs aims to be CO2-neutral for scope 1 and 2'.

Continuous Improvement

The CO2 reduction process is based on the PDCA system: 'Plan - Do - Check - Act' (see figure 4.1). This continuously repeating process is aimed at continuous improvement. With it, we aim to ensure systematic CO2 reduction.

Figure 4.1 PDCA system, source SKAO

4.2 Reduction Targets for Scope 1 and 2

The target is based on our CO2 footprint for 2019. The ambition for 2024 is a reduction of at least 25,5% in Scope 1 & 2 compared with 2019. The tables below show the targets per year and the reduction plan for 2024.

Table 4.1 Targets per year

Year	Scope 1	Scope 2	Scope 1 & 2
2019	reference	reference	reference
2020	5,0%	5,0%	5,0%
2021	7,5%	7,5%	7,5%
2022	20,5%	20,5%	20,5%
2023	23,0%	23,0%	23,0%
2024	25,5%	25,5%	25,5%
2025	28%	28%	28%

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Table 4.2: Reduction results 2024

| Component | Reduction target 2024 vs. 2019 | Emissions 2019 (tonnes) | Emissions 2024 (tonnes) | Already reduced vs. 2019 | Reduction target 2025 vs. 2019 (tonnes CO2) | Reduction target 2025 vs. 2019 (%) |

|---|---|---|---|---|---|---|

| **Scope 1** | | | | | | |

| Reducing emissions from business diesel consumption | 31% | 89,8 | 104,4 | -14,6 | -16% | -5% |

| Reducing emissions from business petrol consumption | 20% | 32,8 | 11,8 | 21,0 | 64% | 20% |

| Replacing outboard motors with electric alternatives | 73% | 1,4 | 0,4 | 1,0 | 69% | 73% |

| **Total scope 1** | 28,6% | 124 | 116,7 | 7,3 | -8% | -2% |

| **Scope 2** | | | | | | |

| Reducing electricity consumption per m² with solar panels | 25% | 25,8 | 21,3 | 4,5 | 17% | 25% |

| **Total scope 2** | 25% | 25,8 | 21,3 | 4,51 | 17% | 25% |

| **Total scope 1 & 2** | 27% | 150 | 138 | 12 | -4% | 2% |

*Total percentages are weighted averages.

Table 4.2 sets out the most significant emission sources for each scope. These sources offer the greatest scope for reducing our emissions. The reduction in emissions is shown for each component. The measures to achieve this reduction are included in the energy management plan in chapter 5.

In February 2025, the targets and results were assessed and new targets were set. For the time being, a total reduction of at least 2.5% per year appears achievable.

Reflection on reduction targets within scope 1

Reducing emissions from business diesel consumption.**

Up to and including 2024, the last old field service vans were replaced with the most sustainable diesel vehicles currently on the market. In 2025, a successful trial demonstrated that an electric alternative can be used for fieldwork. From mid-2025, outdated field service vans will be replaced with electric field service vans. This will reduce our diesel consumption and deliver a visible CO2 reduction.

Reducing emissions from business petrol consumption.**

Over the past few years we have gained more and more clients and more and more work, and fewer people have worked from home than in previous years. The number of kilometres driven has therefore increased. The electrification of our car fleet is progressing steadily, so we hope to emit as much next year in this category as we did in 2019. All business lease cars whose contracts expire will be replaced with an electric alternative. The electric cars will be charged with Greenchoice passes, so that our lease cars run as green as possible. Most petrol cars had already been replaced by 2025. By the end of 2025, there will be a visible reduction in business petrol consumption.

● Outboard motors have been replaced with an electric alternative. Fuel-powered outboard motors are only used when power is needed, such as for many aquatic plants. Given their consumption, the fuel-powered motors are used only incidentally, so further reduction is not possible.

Reflection on objectives within scope 2

● Reducing electricity consumption per m² through solar panels

Our electricity consumption is rising because we are electrifying our vehicle fleet. In 2023 we therefore invested in additional solar panels on the roof of our premises in Amersfoort. We hope to maintain a reduction of approximately 25% compared with 2019 in 2024 as well. Further reduction through solar panels on our own buildings is no longer an option after 2024, as 2024 will be the first year with the maximum number of solar panels. We have, however, chosen Greenchoice charging cards and new charging points at the office, so that Dutch green electricity is used.

Future plan

As already noted in the reflection above, the maximum reduction has been achieved for some of the objectives (outboard motors), or is within sight (solar panels). Given the increase in the volume of work carried out, it should also be noted that emissions are higher, although they may be relatively lower in relation to turnover. We have chosen to charge the vehicle fleet with Dutch green electricity, to replace the charging points, and to trial electric buses and to purchase new ones. The reduction will become visible at the end of 2025, so new objectives per scope will be established at the beginning of 2026.

4.3 Reducing CO2 in our supply chain

At Tijhuis Ingenieurs, the transport of dredging spoil (per axle), the hire of field staff and commuting are the most material emissions within scope 3. This chapter elaborates on these three components. For the transport of dredging spoil (per axle), which accounts for by far the largest emissions in our chain, a summary of the emissions caused in 2024 is also provided.

Commuting

The commuting of our employees causes relatively large CO2 emissions. Through the commuting allowance and expense claims, we have a fairly accurate picture of the kilometres driven in private cars. Under standard rules, commuting falls within scope 3, but we seek to address it just as much as the emissions within scope 1 and 2. The measures we take regarding transport in scope 1 and 2 therefore also apply to commuting. Our objective for commuting is thus the same as our objective for business travel.

Hire of field staff

The CO2 emitted by our hired field staff when driving to our worksites is significant. We make these emissions as transparent as possible. Through the invoices of external parties, we can see how much work they carry out for us. Using average productivity and the average daily driving distance (based on our own field team), we estimate how many kilometres they drive for us. Unfortunately, we have little influence over the sustainable choices made by external parties. The objective is to give preference, when selecting external workers, to self-employed persons (zzp-ers) and companies that place sustainability high on their agenda.

Transport of dredged material

In the execution of dredging work, we have the greatest influence on CO2 emissions during the transport of dredged material by road (per axle). Examples include encouraging the processing of dredged material on adjacent plots, reducing the average transport distance, using sustainable fuels, and other initiatives. Ultimately, we want to be CO2 neutral by 2030. Table 4.3 summarises the emissions for 2020 to 2024.

The report "Rapport ketenanalyse 2024" (Chain Analysis Report 2024) describes the results and the new targets in detail. The summary below outlines these.

To achieve the desired reduction of 40%, we have formulated the following sub-targets that contribute to the desired result:

The use of HVO biodiesel instead of regular diesel.

Reducing the average transport distance.

Making CO2 emissions visible as standard in the WIT multi-year plan.

In the coming years, the above targets must become a matter of course, and we remain open to even more sustainable solutions.

Transport of dredged material: summary 2024

By steering towards spreading dredged material on the banks, 267 tonnes of CO2 were saved on the transport of dredged material by lorry.

The target was to reduce emissions (scope 1+2) in 2024 by 40% during project execution compared with 2021, which amounts to a reduction of approximately 60 tonnes of CO2. Table 4.3 summarises the results of the calculation.

Table 4.3 Emissions of PMC activity "Transport per axle"

Year	Dredged m³	Local processing %	On the banks %	Sustainable fuel (transport) %	Average transport distance	Emissions per m³ (kg CO2)	Total emissions (tonnes CO2)	Reduction vs regular fuel (tonnes CO2)
2020	287.439	63	37	49	1,23	354	0	
2021	303.623	30	70	51	2,46	748	160	
2022	405.317	50	50	49	1,62	657	26,8	
2023	709.192	35	65	44	1,9	1335	20,6	
2024	830.529	16	84	44	2,16	1.789	115	

The following should be noted regarding the table:

In 2024, considerably more work was carried out than in 2023.

In 2024, sustainable fuels were used in 11 projects, which is 6 more than in 2023. This saved 115 tonnes of CO2.

In 2024, we carried out 10 projects with sustainability as an award criterion. These projects emitted 20,4 tonnes of CO2. The projects with an award advantage, including the calculated emissions, can

be found in Appendix 3.

For the sub-target "relatively more dredged material processed on adjacent banks", the ratio in 2024 is much lower than in 2023.

The target "Reducing the average transport distance" was not achieved in 2024; however, it has not got worse either.

Conclusion 2024

Through the use of HVO fuel, we achieved a reduction of 115 tonnes of CO2. In addition, approximately 285 tonnes were spread rather than removed, meaning approximately 285 tonnes of CO2 were saved. This means that our scope 1 & 2 emissions are well below the reduction achieved in our projects.

4.4 Conclusion on achieving targets

Table 4.4 Conclusion on achieving targets

| Emissions | 2019 | Target for 2024 | Emissions 2024 | Reduction | Achieved? | Target for 2025 |

|---|---|---|---|---|---|---|

| Scope 1 | 166.4 | 25.5% reduction | 128.0 | 23.1% | No | 28% reduction |

| Scope 2 | 25.8 | 25.5% reduction | 21.3 | 17.5% | No | 28% reduction |

| Scope 1 & 2 | 192.2 | 25.5% reduction | 149.3 | 22.3% | No | 28% reduction |

| Scope 3 | 435 | 100% reduction of scope 1 and 2 in projects (149 tonnes CO2) | 2287 | 1718 tonnes CO2 | Yes | 100% reduction of scope 1 and 2 in projects (? tonnes CO2) |

4.5 CO2 sample analyses

The laboratory (scope 3) emits 803 tonnes of CO2 per year for the analysis of "our" samples. The laboratory states that it has the ambition to be fully climate-neutral by 2025. It is currently certified under ISO 14001, which supports sustainable business practices. Reasonably speaking, there are no opportunities for Tijhuis Ingenieurs to achieve reductions, as we have no influence over the choices of other companies. Therefore, no target has been formulated for this component (yet).

During the selection round for choosing a supplier (laboratory), the level of CO2 emissions and the laboratory's ambition were taken into account in the assessment. Going forward, we will include CO2 emissions in the supplier assessment for all critical suppliers. We remain in ongoing dialogue with the laboratory about sustainability improvements in areas such as logistics and packaging (see table 5.3).

4.6 Renewable energy

The most significant scope 2 emission source is the electricity consumption of our offices. By applying renewable energy sources, we aim to reduce this. In January 2021, solar panels were installed on the roof of our office in Amersfoort, and in the second half of 2023 the same number of solar panels was added. Tijhuis Ingenieurs shares the building with other companies and holds a 40% share in it. This resulted in 19,200 kWh of renewably generated energy in 2021, equivalent to a saving of 10 tonnes of CO2. In 2022, the yield from the solar panels was 17,400 kWh, equivalent to a saving of 9.1 tonnes of CO2. In 2023, the yield was 19,000 kWh, equivalent to a saving of 9.9 tonnes of CO2. Solar panels were also installed by the landlord at the Hoorn office in 2022, with an estimated saving of approximately 6.8 tonnes of CO2. In 2023, additional panels were installed in Amersfoort. Together with the yield from the extra solar panels in Amersfoort, a total of 21969 kWh of green electricity was generated in 2024.

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4.7 Green Driving

Making the Company Car Fleet Sustainable

In January 2022, the 'Leaseplan Tijhuis Ingenieurs' was launched. This is a first step towards making electric driving the norm. From now on, all lease cars that need to be replaced will be replaced by electric variants over the next three to five years.

Making Field Service Vans Sustainable

Replacing diesel field service vans with electric field service vans is a technical challenge. Our field service vans are heavily loaded, and field workers drive many kilometres per day. Because Tijhuis Ingenieurs likes to pioneer in the field of sustainability, we decided to start a trial with an electric field service van in 2025 nevertheless. The trial was a great success at the beginning of 2025, which is why the Board of Directors chose to order several vans for late 2025 and early 2026.

Electric Pool Cars: Reducing Business Journeys by Private Cars

In February 2023, it was announced that Tijhuis Ingenieurs would take possession of two electric pool cars in August 2023. In the end, both cars were already delivered in June 2023. There is now an electric car at both the Hoorn office and the Amersfoort office. These shared cars may be used by all employees for business journeys and for travelling between the two offices. In this way, we limit business journeys by private cars, which are not always very fuel-efficient. With this, we can make business travel more sustainable and reduce CO2 emissions in this area. This is *the* solution to the high share of business travel in our CO2 emissions at present.

Sustainable Charging Cards

For both the electric lease cars and the electric pool cars, Tijhuis Ingenieurs has purchased charging cards from a sustainable energy supplier, which allow Dutch green electricity to be charged. As a result, since 2024 we have been certain that all cars are charged on the go with genuinely sustainable energy.

5 Energy Management Plan

The most cost-effective way to reduce CO2 emissions is to save energy. This can be achieved by using more efficient equipment and more economical vehicles. Where this is not possible, the problem can be addressed by using sustainable energy and fuels.

The CO2 Performance Ladder works with three scopes:

Scope 1:** All emissions from gas (for example, gas boilers, combined heat and power installations and furnaces) and fuel for the company's own vehicle fleet and plant.

Scope 2:** All emissions arising from the generation of electricity used by the organisation, emissions from business flights, and emissions from business kilometres driven in private cars.

Scope 3:** All other indirect emissions.

This chapter discusses the emission reduction measures we can take. A distinction has been made between the different scopes and ambitions. We have also included measures that have already been taken, separate from the Performance Ladder.

5.1 Measures for CO2 Reduction

The measures are taken from the list drawn up by the SKAO. Measures for the three scopes have been selected from the Milieubarometer. The list is included as Appendix 2. The progress and effectiveness of the measures are reviewed annually during the management review.

Table 5.1 Scope 1 – Fuel

| Measure | Already achieved | Planned | To be investigated |

|---|---|---|---|

| **Business travel:** Charging points for electric vehicles | Two spaces are available at both the Amersfoort and Hoorn offices. The lease plan offers options for home charging points. | | How do we guarantee that green electricity is used for charging at the offices? |

| Green charging cards | Charging cards for sustainable energy have been purchased. | | Continuously monitor how sustainable the charging cards are and whether more sustainable alternatives exist. |

| Electric lease cars | 10 employees drive a fully electric car. | Each new lease car will (reasonably) be an electric variant. | How do we ensure that we drive electrically, as efficiently and as much as possible, as quickly as possible? Which cars? Who will be the first to receive an electric car? |

| Electric shared cars | Electric shared cars are available: one at the Amersfoort office and one at the Hoorn office. | Encourage the use of shared cars. | |

| Electric field service vans | An electric field service van was put into use in 2025. | Each new field service van will (reasonably) be an electric variant. | Monitor developments in electric vans (battery range). |

| Clean/economical delivery vans | All field service vans are recent models, fitted with soot filters and AdBlue engines. | | Where can the clean vans refuel with HVO, and how do we ensure this? |

| Measure | Already achieved | Planned | To be investigated |

|---|---|---|---|

| Tyre pressure | Check the tyre pressure of buses and leased cars every 3 months. | Check the tyre pressure of buses and leased cars every 3 months. | |

| Fuel-saving equipment | Field service buses: cruise control; gear-shift advice; start-stop system on new buses. | | |

| Energy-efficient driving style, "new driving" | We have a young field service team. They learned this while obtaining their driving licences. | Discuss fuel saving and raise awareness during field service meetings. | Online education module? |

| Choose the best tyres (rolling resistance) | The leasing company handles maintenance for the buses and leased cars. | Make careful trade-offs between cost and "fuel-efficient" tyres when purchasing. | Can we choose tyres with lower rolling resistance? What benefit can we achieve in CO2 reduction and costs? |

| Promote cycling | Since late 2023, we have offered a cycling allowance equal to the mileage allowance for commuting car trips to all colleagues who cycle between 0 and 30 km. Since 2025, there has been a bicycle plan, through which employees can purchase a bicycle at a favourable price. | | |

| Gas & heat | Optimising heating control (start-up) | Optimise heating control (start-up) | Optimising heating control (start-up) |

| Electric heating | | Electric heating via the air-conditioning system in Amersfoort, to become gas-free. | Find out whether this is also possible in Hoorn (possibly in a new building). |

| Improve insulation | Improve insulation | Improve insulation | Improve insulation |

| Time switch for space heating | Time switch for space heating | Time switch for space heating | Time switch for space heating |

| Improve draught-proofing | Improve draught-proofing | Improve draught-proofing | Improve draught-proofing |

Table 5.2 Scope 2 – Electricity

| Measure | Already achieved | Planned | To be investigated |

|---|---|---|---|

| Electricity: sustainable energy contracts | | For both the Hoorn office and the Amersfoort office, it is being investigated whether sustainable | |

Measure | Already achieved | Planned | To be investigated

Energy contracts are, or can be, concluded by the landlord.

Switch off equipment outside working hours.

(Future) electric cars

Use charging passes with which electric cars are always charged with genuine Dutch green electricity.

Continuous monitoring that charging passes supply Dutch green electricity.

Solar panels (PV panels)

In 2021, solar panels were installed on the roof of the Amersfoort office. Tijhuis Ingenieurs holds a 40% share. In 2021 this amounted to 7.501 KwH (4.2 tonnes CO2). Halfway through 2023, more solar panels were installed in Amersfoort. In 2024 we will be able to read the annual yield from these.

In 2022, 615 solar panels were installed by the landlord of our building in Hoorn. We use 6% of the building, which amounts to roughly 36 solar panels. This generates 12.789 KwH per year (6.2 tonnes CO2).

Are there options for more accurately quantifying the solar power generated in Hoorn?

Hold more digital meetings, both internal and external

In 2020, most meetings took place digitally. As a result, the use of electricity and fuel was reduced. Employees are now given the opportunity to work from home one to two days a week, which leads to a reduction in commuting traffic.

We encourage clients to hold interim and short meetings online.

Keep interim external project meetings digital. Start-up meetings continue to follow the regular method.

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Measure | Already implemented | Planned | To be investigated

Energy-efficient electric motor

All field teams have an electric outboard motor. Where possible, it is used. In addition to CO2 reduction, this also has benefits for working conditions.

Investigate how the electric outboard motors can last as long as possible.

Capacity control of room ventilation

The air conditioning at the Amersfoort office (not present in Hoorn) is adjusted based on the occupancy rate of the room in the building.

Switch lighting per (part of) room

In the offices, lighting can be switched on per room or segment. Given the occupancy rate of the offices, it seems unnecessary to further divide the segments with switches.

Room empty: light off via sensor

Main lighting is switched off at the end of the day. Spotlights switch off automatically.

Replace fluorescent lighting with lighting using starters

Fluorescent tubes with electronic ballasts, or LED tubes, to be purchased for the Hoorn and Amersfoort offices.

Replace halogen lamp

Replace with LED lighting

LED emergency lighting

Energy-efficient outdoor lighting (replace halogen and high-pressure mercury lamps)

Amersfoort: replace with LED lamps

Desktops, laptops and monitors with Energy Star

Laptops from 2019 onwards carry this certification. Older laptops have now almost all been replaced.

Process: continue to give priority to equipment with low standby consumption or a more favourable energy label

Centralised printing and copying

Measure | Already met | Planned | To be investigated

Annual efficient management and maintenance of (recognised) energy measures

Each year, assess whether the planned measures are functional, whether they are being adhered to, and what the progress of implemented measures is.

New toilets

The new toilets at the Amersfoort office save a great deal of water, using a small and a large flush button.

Table 5.3: Scope 3 – Advice & transport of dredged material

Measure | Already met | Planned | To be investigated

Execution projects

Advice: Transport of dredged material per axle (distance and fuel)

Objective partially achieved:

• Use sustainable fuels in 5 projects

• Sustainability as an award criterion in 12 projects

Reduce our own scope 1 & 2 emissions by 40% during execution projects, through:

• Method of processing (on-site treatment or removal)

• Transport distance (choosing a processing location nearby)

• Sustainable fuels (specifying HVO instead of diesel)

Do we have sufficient influence over "on-site treatment" and transport distance? Or does our influence lie precisely in the type of fuel or the equipment deployed?

Reduce our complete scope 1 & 2 emissions during execution projects within 5 years.

Reuse of sample pots

Remain in discussion with Eurofins Analytico to encourage and investigate the reuse of packaging.

Keep an eye on the available options for reusing packaging.

Electrifying logistics Analytico

Analytico has pledged its willingness to work with us.

Pilot with Analytico to trial an electric field-work van at Tijhuis Ingenieurs, in order to reduce transport emissions.

Optimising logistics (Eurofins Analytico)

What are the options for driving a more efficient route in collaboration with Eurofins Analytico and other companies?

Scope 3 Measures

To measure and achieve this target more precisely, the points below are applied. Various examples are given under each component.

Create awareness and understanding (internally and externally) by:

gaining greater insight into the transport choices and options of contractors, through (source) research;

creating broad internal awareness of the opportunities for CO2 emission reduction in the transport chain. This concerns knowledge of cleaner fuels for transport equipment, knowledge of techniques for reducing the volume of process water, and so on;

creating awareness among our clients in the preparation phase of CO2 reduction opportunities in the transport chain. Once the scope of a project is known, we prepare a sustainability report.

Make objectives and measures measurable by:

drawing up sustainability reports.

Once the scope of a project is known, we prepare a sustainability report. In it, we calculate the CO2 emissions and set out the reduction opportunities in the transport chain. A calculation template is also added, with which the impact of the CO2 reduction measure in the transport chain can be calculated directly. The final choice is then taken into account later in the specification;

drawing up EMVI criteria for sustainability (within the sub-targets). The client must reward these alternatives with a significant (notional) discount;

making current CO2 emission reduction measures of the hauliers transparent, so that we can then steer more explicitly towards points for improvement and make a comparison between 2020 and the future.

Make our way of working more sustainable by:

specifying low-CO2 or zero-CO2 alternatives, such as hybrid lorries, hybrid cranes, or lorries that use cleaner fuels;

innovating to reduce transport movements.

As a company, we are looking into methods of reducing the number of transport movements of dredged material. Forward-looking solutions are being mapped out in order to bring the importance of this type of solution to the client's attention at an early stage of a project. An example of this is thickening sludge through the use of flocculant, which greatly reduces process water and means a smaller volume of dredged material needs to be removed. This results in considerably fewer transport movements for the same quantity of dredged material.

Progress on reduction targets and measures is reported every six months in internal communications. In the next version of the CO2 Handbook (2025), these measures will be updated again.

6 Chain Initiatives

This chapter discusses our active participation in chain initiatives. These initiatives are set up to make both our own organisation and the sector aware of the opportunities and benefits of CO2 reduction.

6.1 Possible Initiatives

We see the following opportunities:

providing input during meetings;

setting up innovations;

CO2-conscious procurement.

An annual budget of at least € 2.500 excluding VAT has been set for participating in and setting up initiatives. This is included as a fixed item in the annual budget.

We are always actively looking for chain initiatives to join, and we explore the possibilities of setting them up ourselves. Section 6.2 lists the ongoing initiatives, including their potential impact on CO2 reduction in the chain.

6.2 Ongoing Initiatives

Automating Our Work Processes

Over recent years we have invested heavily in automating our work processes. As a result, far less is printed, and field workers no longer need to go to the office every day to collect and deliver their work. This has led to a significant CO2 reduction (this is < 2019 and not quantified).

We have developed the following applications, which are relevant to CO2 reduction:

WIT Metingen;

WIT Meerjarenplanning;

WIT Data;

WIT Uitvoering;

WIT Waterpaspoort.

We also offer these applications to our clients so that they too can contribute to reducing CO2 emissions in the chain.

Sustainable Supplier

'Duurzame Leverancier' is a knowledge platform for companies and organisations that wish to operate safely and responsibly. The platform offers the opportunity to take part in sounding board meetings and to exchange knowledge and information. We are a member of this platform.

On 4 February 2024, Duurzame Leverancier published an interview with one of our directors. The article is available online at:
duurzameleverancier.nl/blog/81/In-the-spotlight:-Myrte-Tijhuis,-eigenaar-van-Tijhuis-Ingenieurs

Membership of SKAO

In the run-up to certification, we became members of SKAO. Through this platform, we can take part in sounding board groups, webinars and other meetings. We actively participate in these. At the very least, the following employees follow an initiative at least once a year: the Director (M.M.A. Tijhuis) and the CO2 project leader (E. van Wijk-Reus). In addition, we give our colleagues the opportunity to take part in these initiatives through an open registration.

Sustainable materials and equipment

We use sustainable materials wherever possible. CO2 is already emitted during the production of every piece of equipment used. By using sustainable materials, CO2 emissions are reduced.

We are taking the following initiatives in this regard:

● Titanium piston auger

We have developed a titanium piston auger. It wears less quickly and is lighter. This piston auger is (in addition to better ergonomics) far more sustainable than the conventional steel variant. Initially, we developed this auger for our own fieldwork team, but it is now also used by various water board laboratories and water boards. We also let other companies benefit from this innovation; several companies have purchased piston augers from us.

● Titanium gauge rods

Just like titanium piston augers, titanium gauge rods are lighter and more sustainable than the conventional variant. The design was created in 2022 and is still being further developed.

● Sustainable fieldwork van equipment

We are also examining the equipment inventory of the fieldwork van to identify plastic tools that can be replaced with a more wear-resistant alternative, for example stainless steel (RVS) buckets instead of plastic ones.

● Electric outboard motors

During fieldwork, an electric motor is always used for the boats. Only if there is genuinely no alternative is a petrol motor used instead.

Circular Dredging Consortium

Since 2025, Myrte has been chair of the Circular Dredging Consortium, a partnership of frontrunners in the sustainable dredging industry. The aim of this organisation is to make the dredging chain circular by supplementing current application possibilities with innovative circular applications. See also Circulair Bagger Consortium | samenwerken voor een circulaire baggerketen.

Stichting de Bloeiende Oevers

At the end of 2024, Stichting de Bloeiende Oevers was established by Hans and Myrte Tijhuis. Alongside the commercial Tijhuis Ingenieurs, they wish to give something back to nature through the foundation, on a non-profit basis. The heart of Stichting de Bloeiende Oevers is blue and green. Its goal is to connect water and greenery in order to make aquatic life more beautiful. Drawing on the experience of Tijhuis Ingenieurs and in collaboration with water managers, municipalities,

nature organisations, landowners, consultancies and contractors, the foundation aims to make the water-rich Netherlands more beautiful and to improve the quality of the water system and biodiversity. See also Stichting de Bloeiende Oevers | LinkedIn.

Planting greenery

We annually support sustainable initiatives with donations, such as Hoop heggen (www.hoopheggen.nl). Planting greenery contributes to a sustainable society. This is not only good for biodiversity. A green outdoor environment also contributes to CO2 reduction. In October 2021, the entire company also spent a day planting hedges together. In total, 400 metres of hedge were planted, which corresponds to

with approximately 6 tonnes of CO2 compensation (Compensation 15 kilograms of CO2 per metre of hedge, source https://hoopheggen.nl/doneer/#nu_doneren).

Figure 7.1: Planting hedges with HoopHeggen (Archive Tijhuis Ingenieus).

PHI Factory

One of our directors, Myrte Tijhuis, was interviewed by PHI Factory. Through this, we hope to inspire other companies to also get started on sustainability. The interview can be found here: Aan de slag met het Waardenwiel bij Tijhuis Ingenieurs (phifactory.com).

CO2 Reduction in Business Travel

In 2018, we had two electric vehicle charging points installed at each of our locations in Hoorn and Amersfoort. This paved the way for greening our vehicle fleet. Currently, 7 of our 16 lease cars are electric. In addition, since June 2023, we have had two electric pool cars. These can be used by all colleagues for business travel and commuting. Our target is for 75% of the fleet to be electric by 2025. For our fieldwork vans, this is a challenge with current technology, given the battery capacity in relation to the distances driven per day. In collaboration with the leasing companies and our laboratory, we are looking for options to green the fleet. We are also seeking solutions to reduce transport distances, for example by discussing this with the laboratory. Furthermore, we are in discussions with the lease car providers and the CO2 emissions reports they supply.

7 Communication – CO2 Performance Ladder

This chapter contains the communication plan. It sets out how we communicate with internal and external target groups about CO2 reduction and our objectives.

Responsibilities

Internal and external communication about the CO2 Performance Ladder and related documents is the responsibility of the CO2 project group, directed by the management team. E. van Wijk-Reus, as project leader, is responsible for keeping the CO2 portfolio up to date. In collaboration with the communications officer (D. Schaafsma), the information is published and maintained on Tijhuis Ingenieurs' own website and social media.

7.1 Target Groups

Internal target group

The internal target groups of Tijhuis Ingenieurs are the management, office staff (consultants, geodata and administration) and the field team. The group comprises approximately 65 people, and our employees are highly involved. Our objective is to make the entire organisation even more aware of our CO2 emissions, reduction targets and sustainable choices in general.

External target groups (stakeholders)

The external target groups are (prospective) clients, contracted staff, partners and (prospective) suppliers. Our most important client is the government (water boards and municipalities). The government considers it important that organisations operate in a socially responsible way. Our objective is to make as many external parties as possible aware of the importance of CO2 reduction. In addition, we are keen to show what we are doing to become more sustainable, for example by reducing our CO2 emissions. Our clients can contribute to sustainability through their procurement policies.

Positioning

Tijhuis Ingenieurs aims to be a pioneer in sustainability within our sector. By achieving level 5 on the CO2 Performance Ladder, we demonstrate that we are contributing to a clean future. Furthermore, through the measures and initiatives we have taken, we want to encourage the sector to reduce its CO2 emissions as well. It is no coincidence that our slogan is:

‘Tijhuis Ingenieurs: sustainable advice in and around water’

7.2 Communication Objectives

Internal objective

All colleagues know how they can contribute to CO2 reduction and sustainability within Tijhuis Ingenieurs. Colleagues also know which steps the company has taken, and they help think through the measures to be taken.

External objective

It is clear to external parties that we work in an environmentally conscious way, because we consider sustainability very important. In addition, we show that socially responsible business practice is high on the agenda at Tijhuis Ingenieurs.

7.3 Communication Plan

Our communication plan sets out how we communicate internally and externally. The table below provides a summary of the communication moments.

Table 7.1 Overview of communication moments

Communication channel	Scheduled	Subject	Responsible party(ies)
Internal communication			
Social media & Teams	Every two months	Update on sustainability, projects or milestones	Marketing & Communications, sustainability team
Quarterly presentations	March	CO2 reduction and targets + measures (previous year's data)*	Management, sustainability team, CO2 coordinator
	June	General sustainability	
	September	CO2 reduction and targets + measures (data for first half of current year)*	
	December	General sustainability	
Start of the month	Monthly	General sustainability	Sustainability team, CO2 coordinator
	March	CO2 reduction and targets + measures (previous year's data)*	
	September	CO2 reduction and targets + measures (data for first half of current year)*	
Keep information up to date on the SKAO website	March	Update after calculating CO2 footprint	CO2 coordinator
	September	Update after calculating CO2 footprint	
External communication			
News items, social media	Every two months	Update on sustainability, projects or milestones	Marketing & Communications, sustainability team
Publication on the website (sustainability page)	March	Update after calculating CO2 footprint + update of 'CO2 Performance Ladder Manual' (previous year's data)	Marketing & Communications, CO2 coordinator
	September	Update after calculating CO2 footprint (data for first half of current year)	
Annual report on projects with award advantage	January	Summary of CO2 footprint and milestones of projects with award advantage (data from last year)	Marketing & Communications, CO2 coordinator

* Communication during the quarterly presentation OR at the start of the month

Mandatory CO2 Performance Ladder Communication

Publication topics

CO2 footprint and reduction targets;

measures taken or being taken;

projects obtained thanks to the CO2 Performance Ladder;

initiatives and ideas from colleagues, clients, subcontractors or suppliers.

Post-Certification Process

This document has been prepared in preparation for our certification for the CO2 Performance Ladder level 5). Once the manual and accompanying documents have been approved by an external certification body (CB), the public documents will be published on the own website of Tijhuis Ingenieurs.

Quarterly presentations and start of the week

Verbal communication is preferred over written communication when we communicate extensively about CO2 and sustainability. In this way, sustainability becomes more of a living concern within the company, making employees more inclined to think about it while working on projects.

Since 2023, sustainability has therefore been given attention at all company-wide quarterly presentations and monthly meetings. In addition, at least one quarterly presentation per year is entirely devoted to sustainability, and regular updates are given on the CO2 Performance Ladder and our emissions. During these verbal communication moments, employees are encouraged to participate and think along. Examples of this include:

Completing a form asking for ideas to improve the sustainability of the company.

Asking the audience (the employees of Tijhuis Ingenieurs) questions during presentations to open a lively discussion.

At the start of the month (monthly meeting), colleagues are regularly made aware of sustainable innovations and achievements, after which there is always room for questions.

8 Assurance

8.1 Internal Audit

Scope / Reach

The internal audit assesses the Handbook and all other documents relating to the Portfolio.

Implementation

The KAM coordinator carries out an internal audit annually in the first quarter. This is conducted in parallel with the other internal audits (ISO/VCA/BRL). This is recorded on the audit form together with the self-evaluation. The results are discussed with the MT and the board of directors, on the basis of which the follow-up steps are established. If shortcomings are identified, objectives are set and deadlines are established. A summary of this analysis is included in the annual management review.

Roles

Board of directors/MT: Responsible for the assessment and follow-up steps;

KAM coordinator: Carries out the internal audit and assesses any follow-up steps;

Auditee: The party being audited;

CO2 project leader: Collects data for the documents required for the CO2 certificate.

Content of the audit report

The audit report records the functioning of the company under the CO2 Performance Ladder. For each requirement of the CO2 Ladder and each internal objective, the following is examined:

● Are we making progress in achieving the objective underlying the requirement?

● Which additional measures must we take to achieve this? Consider not only technical measures, but also awareness, attitude and behaviour.

The results are included as input to the management review and discussed with the board of directors.

8.2 Management Review

The management review is an evaluation of the past year, comparing it with the previous year. Key points are: have the objectives been achieved, where do we see improvements, and how do we achieve them?

Scope / Reach

Concerns all processes relating to the CO2 Performance Ladder within the business operations.

Implementation

The KAM coordinator and the board of directors assess the quality system of the company annually. From now on, the CO2 Performance Ladder will also be included on the basis of the following questions:

● Does the CO2 Performance Ladder function within the company as intended, given our performance against the objectives per requirement?

● Are the existing measures sufficient to achieve the reduction objectives?

● Are the reduction objectives still sufficiently ambitious?

● Is the initiative still sufficiently useful? Does it still align with reduction opportunities? Can we still provide sufficient information in the coming period?

● What can we improve, and which improvement do we tackle first?

Is the collection of data proceeding well, is it of sufficient quality, and do we have enough insight?

Which new and challenging measures can we test in a pilot in the coming period?

Do we want to grow towards a higher level?

Should we adjust the deployment of a measure where monitoring shows that it is not effective?

Have the improvement points from internal and external inspections and audits been addressed and implemented?

Roles

Management / Management Team (Directie/MT): Responsible for the assessment and follow-up steps.

KAM coordinator: Collects the required documents and, together with the management, carries out the management review.

CO2 project leader: Auditee.

8.3 Publications

Purpose

Publishing the CO2 data on the Tijhuis Ingenieurs website and keeping the public documents, such as the CO2 Manual and its related publications and documents, up to date.

Scope

This concerns documents relating to the CO2 Performance Ladder that are publicly available.

Method

Measurement data is recorded periodically, for example meter readings and fuel consumption. Each year we use this data to complete the Milieubarometer (Environmental Barometer) and compare the results to map out CO2 reductions. The Manual and other data are updated and shared publicly on our website. This step is carried out in the first quarter in preparation for the external audit.

Tijhuis Ingenieurs Website

Our website meets, as a minimum, the following requirements:

It can be reached under the name of the organisation (as stated on the certificate) and then via the term "CO2 Performance Ladder" or "CO2 policy".

It contains at least the required information (and documentation) as set out in the explanatory notes to the requirements of the SKAO CO2 Manual. This information remains available online for at least the duration of the certificate, with a minimum of 2 years.

The organisation refers, with a clear link, to the organisation page on the SKAO website where the documents for requirements 4.A.1, 3.D.1, 4.D.1 and 5.D.1 can be found.

The organisation's website carries a full copy of the valid certificate.

If there is a change in requirements 1 to 5 of the CO2 Manual, the website will be updated within 4 weeks of the information being finalised.

Publishing on social media is not considered communication on the organisation's website. Social media publications may be used as a supplement.

Publication of Tijhuis Ingenieurs BV on SKAO

This website can be found at www.skao.nl.

Tijhuis Ingenieurs B.V.

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